Overfitting, regularization, and model validation

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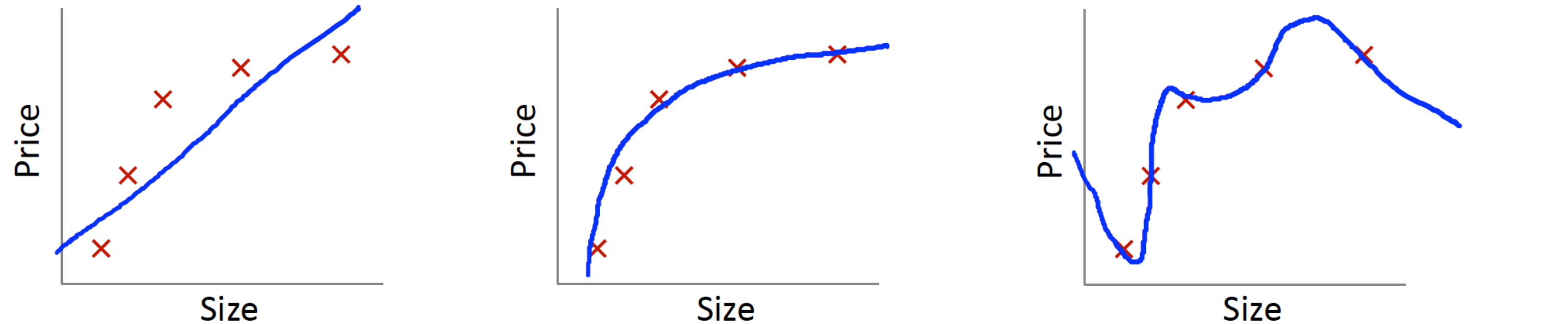
Content

- Overfitting
- Regularization
- Model validation

Overfitting



$\theta_0 + \theta_1 x$ $\qquad$ $\theta_0 + \theta_1 x + \theta_2 x^2$ $\qquad$ $\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

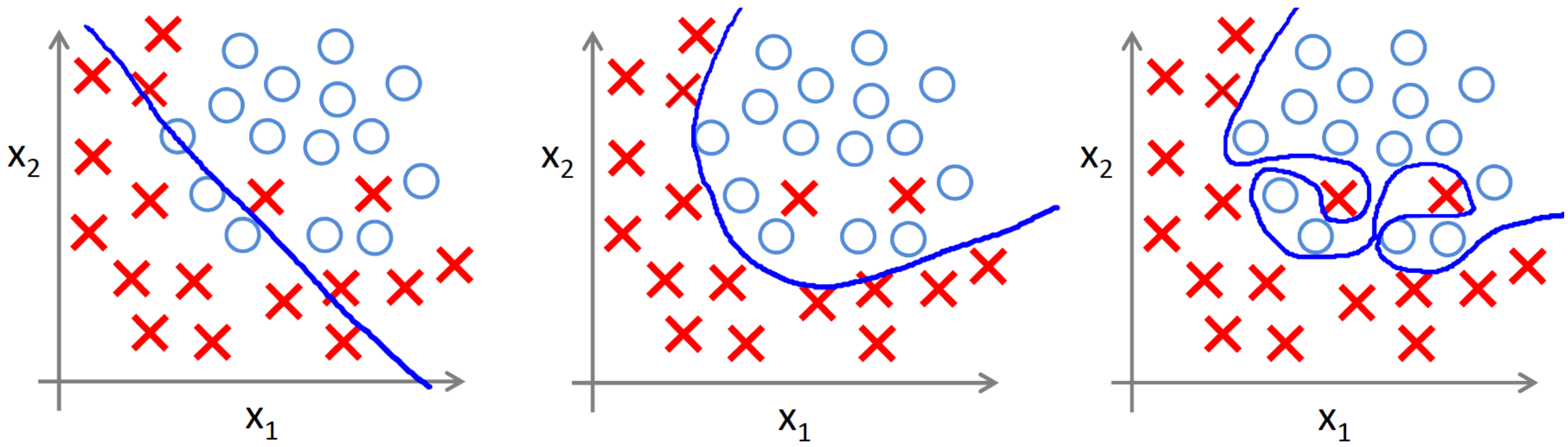
When there are many features:

- Model may overfit with training data (training cost ~ 0)
- Could not generalize for new samples (testing cost >> 0)

Source: Andrew Ng

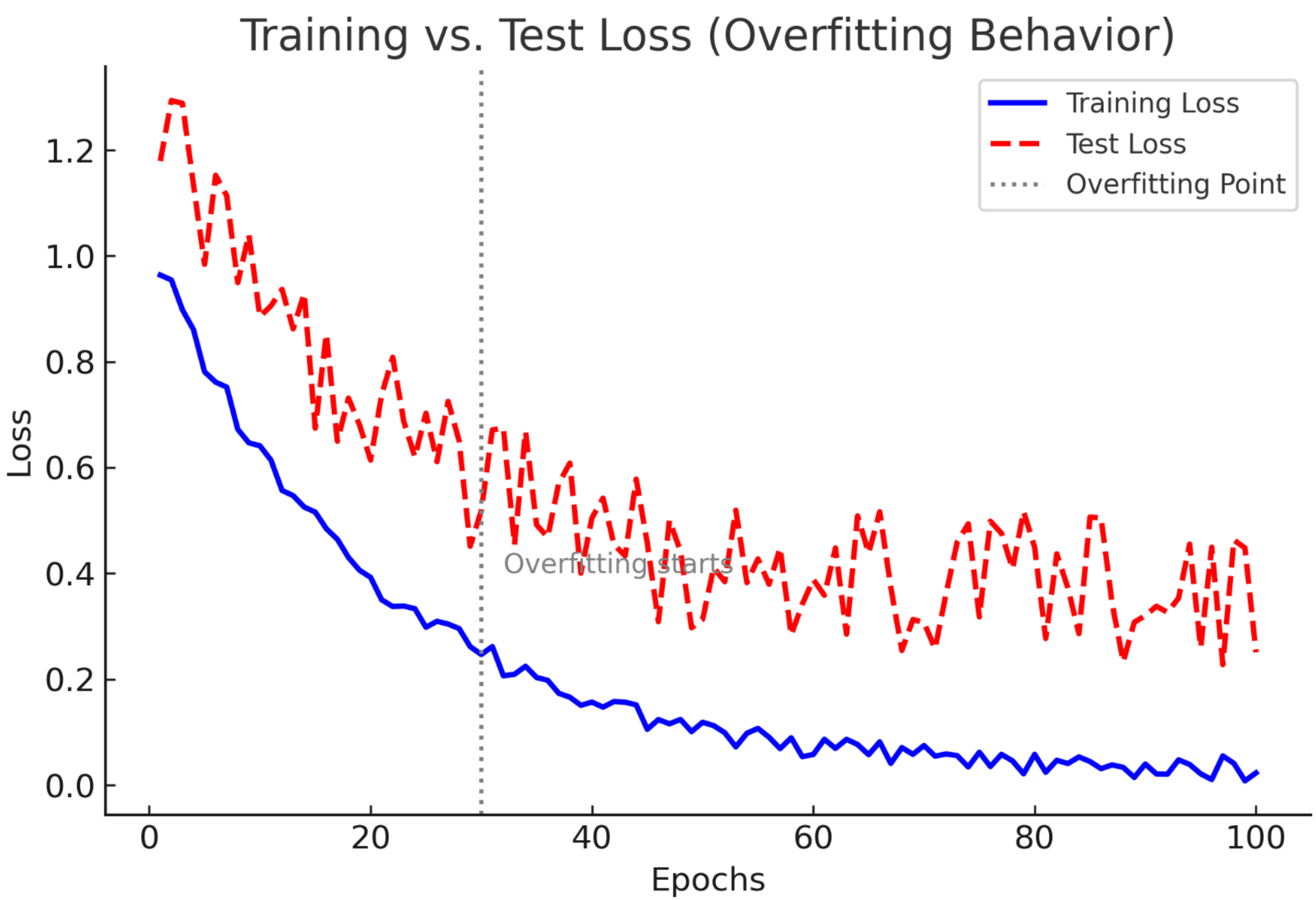
Overfitting

- Logistic regression



Source: Andrew Ng

Overfitting



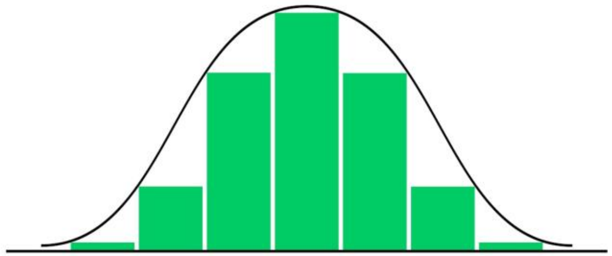
Source: ChatGPT

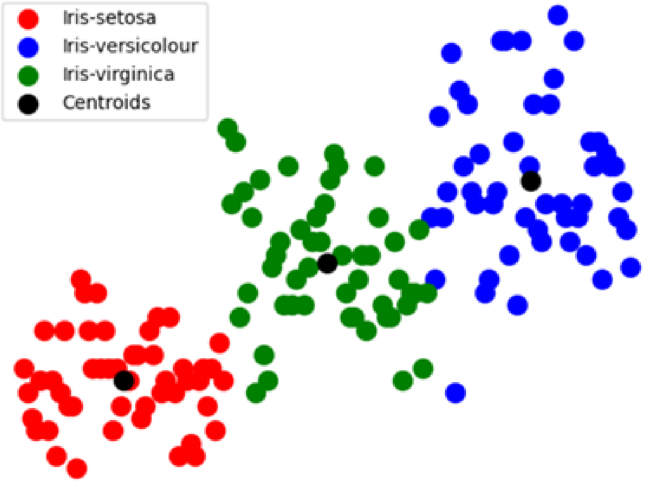
Overfitting

- Solutions:
  - Reduce number of features
    - Select features through data analysis
    - Select features through model validation
  - Regularization
    - Reduce effect of parameters to output
    - Good in case features contributes slightly to output

Feature selection

- Learn structure, distribution and relationship between input and output
  - Tools like histogram or scatter plot may be helpful





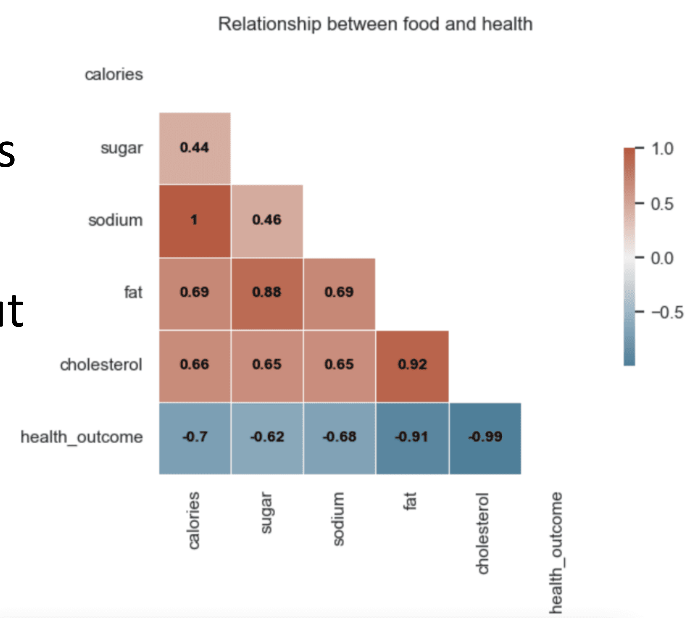
Source: Internet

Feature selection

- Evaluate importance of feature using correlation
  - Apply to linear relationship
  - Determine correlation inside features
  - Determine correlation between each feature and output
- Feature that is highly correlated with others can be eliminated
- Feature that is highly correlated with output is more important

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient
x_i = values of the x-variable in a sample
$\bar{x}$ = mean of the values of the x-variable
y_i = values of the y-variable in a sample
$\bar{y}$ = mean of the values of the y-variable

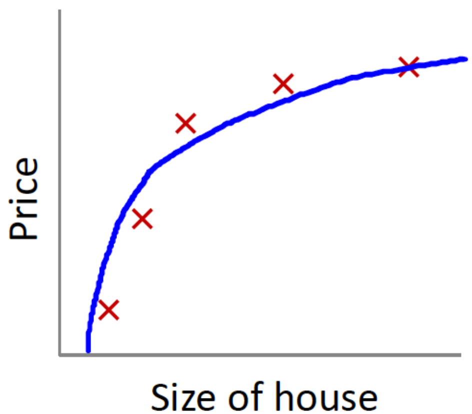


Source: Internet

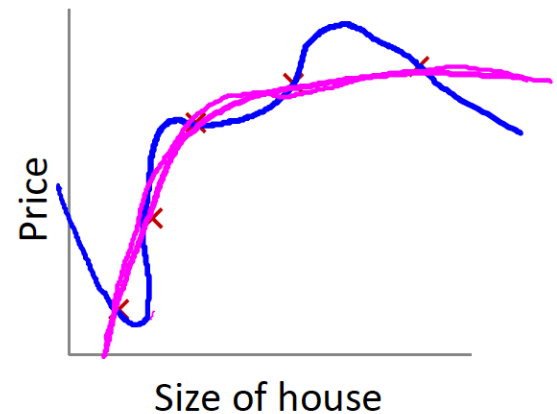
Feature selection

- Steps to select features
  - Learn structure, distribution and relationship between input and output
  - Evaluate importance of features using correlation
- Model evaluation
  - Test model with selected features
  - Repeat feature selection until model is good enough

Regularization



$$\theta_0 + \theta_1 x + \theta_2 x^2$$



$$\underline{\theta_0 + \theta_1 x + \theta_2 x^2} + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

$$\min_\theta \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 + 1000\theta_3^2 + 1000\theta_4^2$$

Source: Andrew Ng

Regularization

- When values of parameters become small
  - Produce simpler models
  - Reduce overfitting

Without regularization: the model overfits by assigning large weights to features, leading to high variance.

With regularization: the model penalizes large weights, leading to better generalization.

L2 Regularization

- Cost function

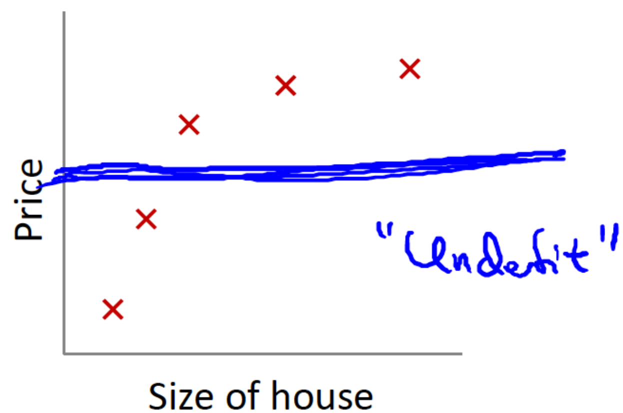
$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 + \lambda\sum_{j=1}^{n}\theta_j^2\right]$$

- Lambda: weight decay
- When lambda is bigger, model becomes simpler

L2 Regularization

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$

If lambda too big?

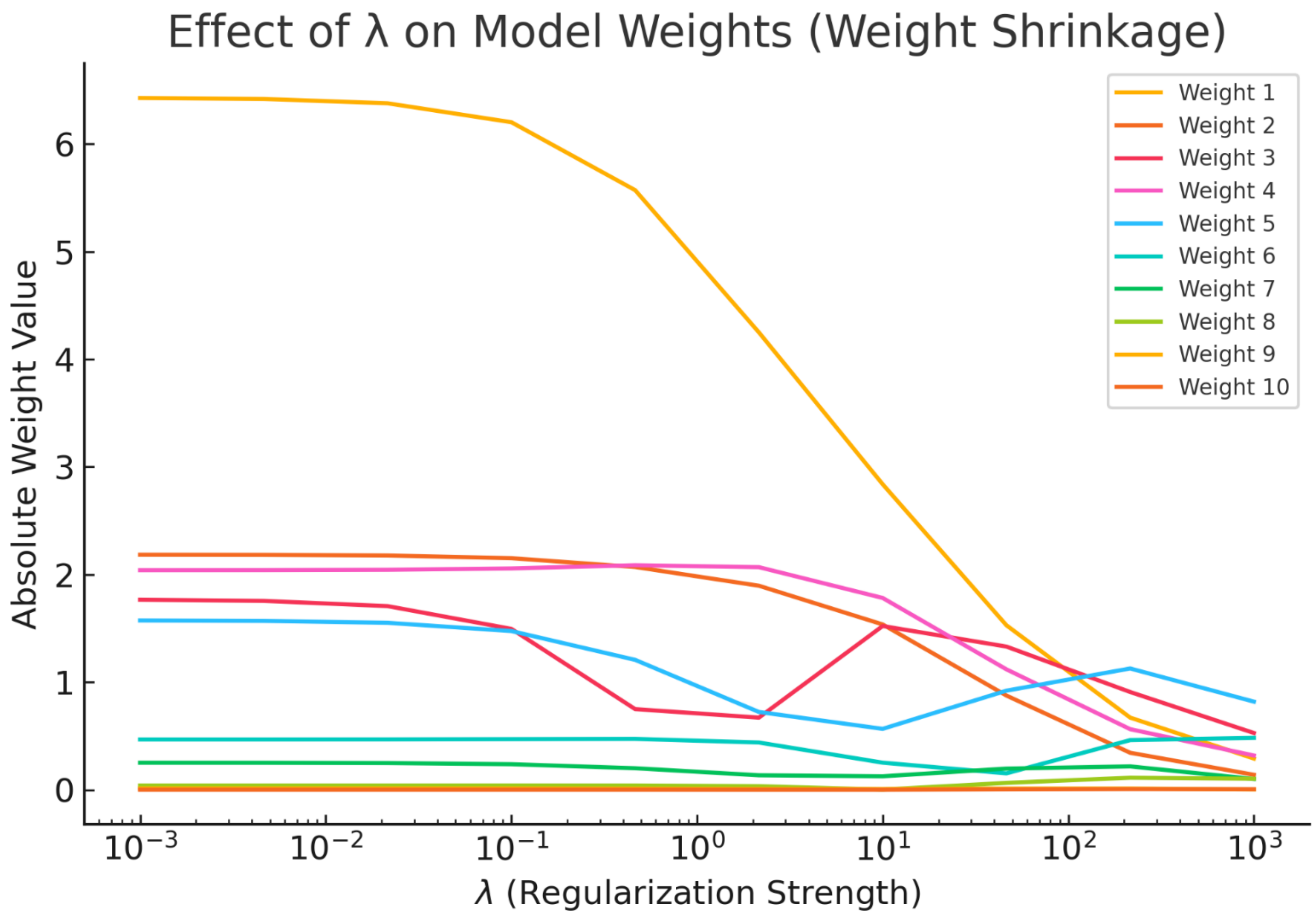


$h_\theta(x)$

$$\theta_0 + \cancel{\theta_1 x} + \cancel{\theta_2 x^2} + \cancel{\theta_3 x^3} + \cancel{\theta_4 x^4}$$

Source: Andrew Ng

L2 Regularization



Source: ChatGPT

L2 Regularization

- Vector gradient

$$\frac{dJ}{d\theta_0} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)x_0^{(i)}$$

$$\frac{dJ}{d\theta_j} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)x_j^{(i)} + \frac{\lambda}{m}\theta_j$$

Gradient descent algorithm works as usual

L2 Regularization

- Logistic regression – cost function

$$J(\theta) = -\left[\frac{1}{m}\sum_{i=1}^{m} y^{(i)}\log h_\theta(x^{(i)}) + (1-y^{(i)})\log\left(1-h_\theta(x^{(i)})\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$$

L2 Regularization

- Logistic regression – vector gradient

$$\frac{dJ}{d\theta_0} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)x_0^{(i)}$$

$$\frac{dJ}{d\theta_j} = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)x_j^{(i)} + \frac{\lambda}{m}\theta_j$$

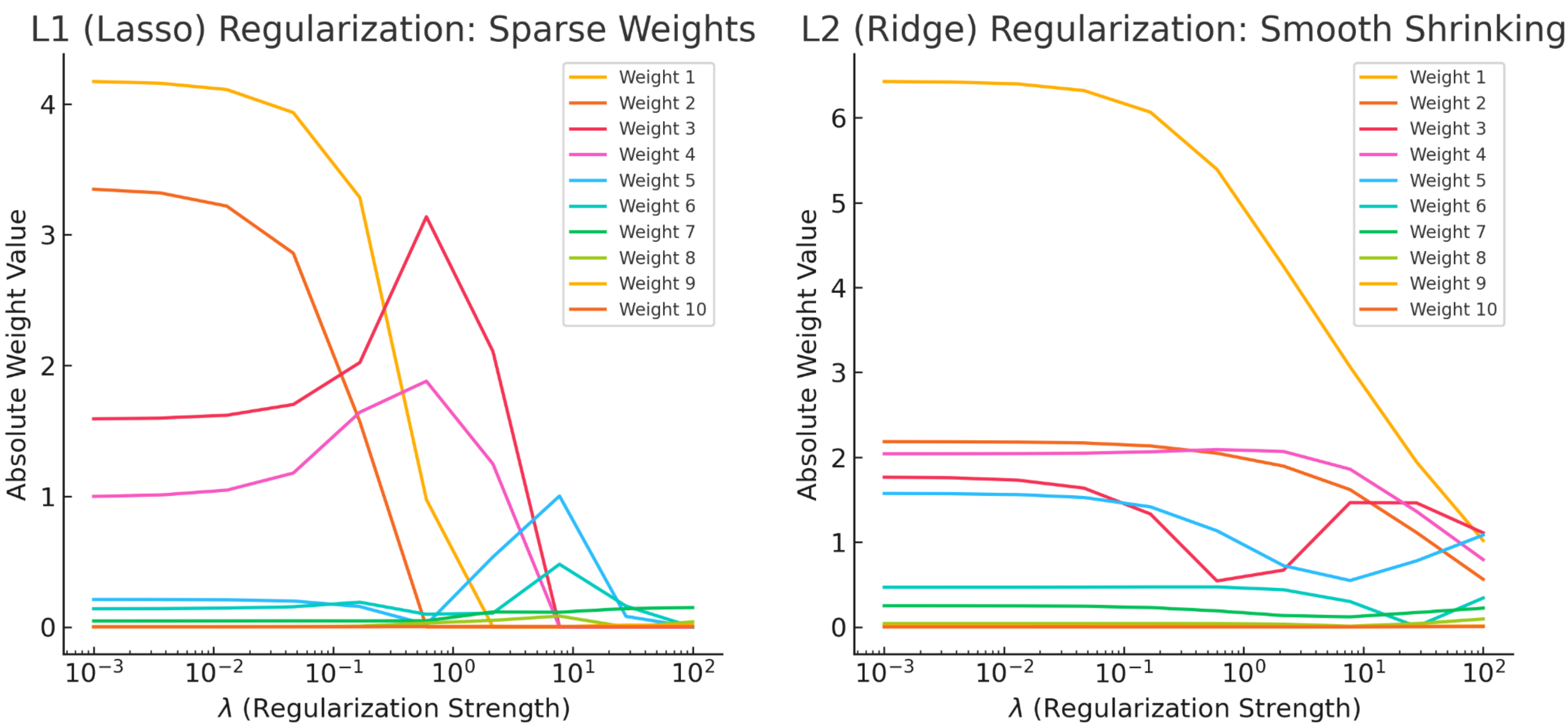
L1 Regularization

- Cost function

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 + \lambda\sum_{j=1}^{n}|\theta_j|\right]$$

- Adds penalty on sum of absolute weights
- Effect: force some weights to become exactly zero
- Perform feature selection by eliminating irrelevant features

L1 an L2 Regularization



Source: ChatGPT

Overfitting

- Other reasons for overfitting:
  - Training set size is too small
  - Training samples do not represent all possible input data
  - Training samples contain large amount of irrelevant information or noisy data
  - Model is overtrained

Overfitting

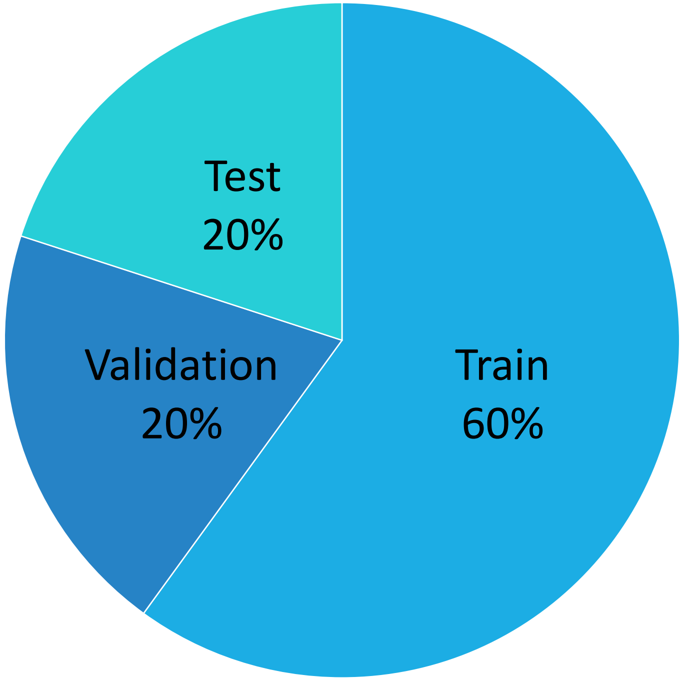
- Other solutions for overfitting:
  - Data augmentation
    - Increase number of training samples by slightly changing available samples
  - Early stopping during training
    - Training is stopped before the model learns noise in data
  - Ensembling
    - Combine predictions from several models to get more accurate results

Model validation

- Learning process makes model fit with training data
- Can learnt model generalize for new samples?
  - Validate the model with unseen data
- Solution:
  - Learn, validate, and select the best model
  - Test if the selected model works well with new data

Dataset

- Split dataset into 3 subsets: train, validation, and test
- With large dataset, the ratio is: 60% : 20% : 20%



Cost function

Train error

$$J_{\text{train}}(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

Cross-validation error

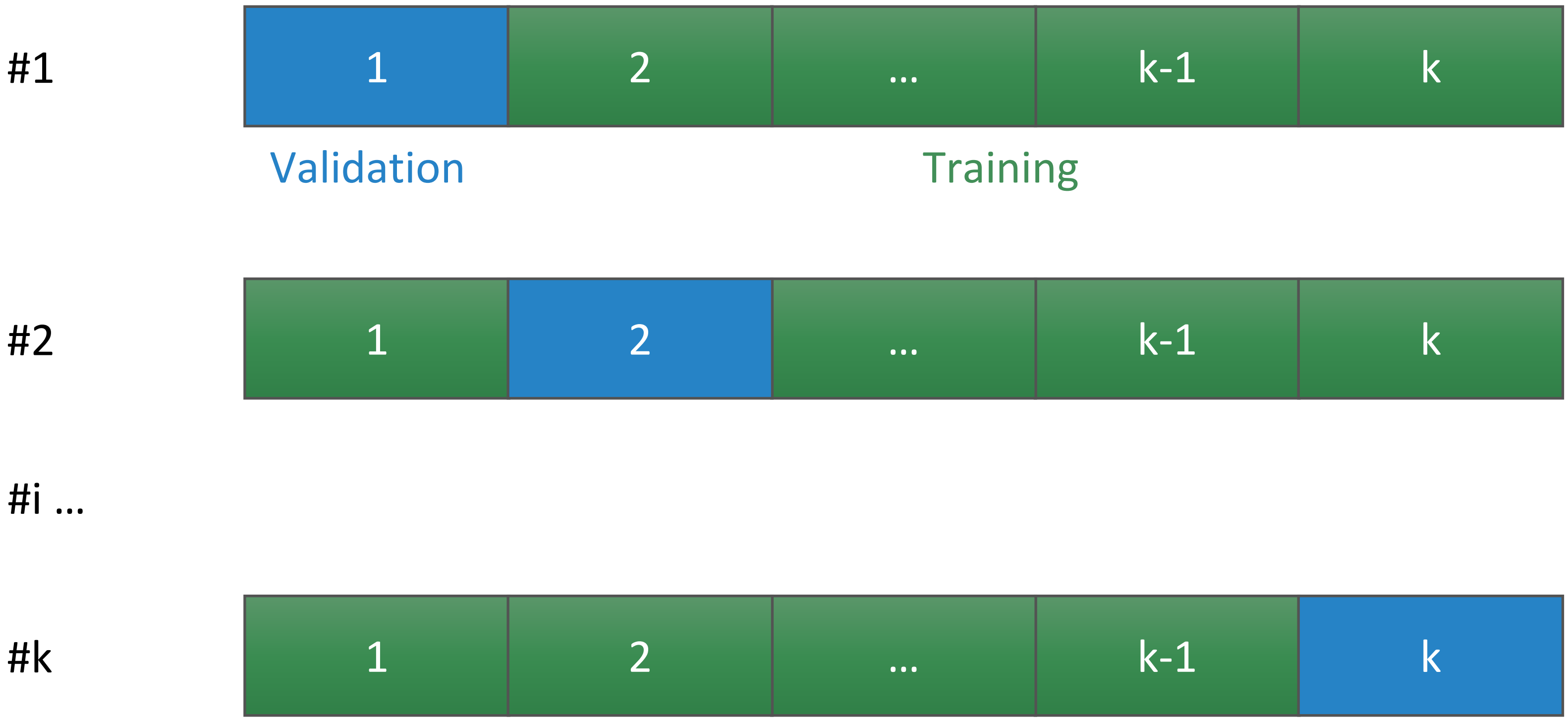
$$J_{\text{cv}}(\theta) = \frac{1}{2m_{\text{cv}}} \sum_{i=1}^{\text{cv}} \left(h_\theta(x_{\text{cv}}^{(i)}) - y_{\text{cv}}^{(i)} \right)^2$$

Test error

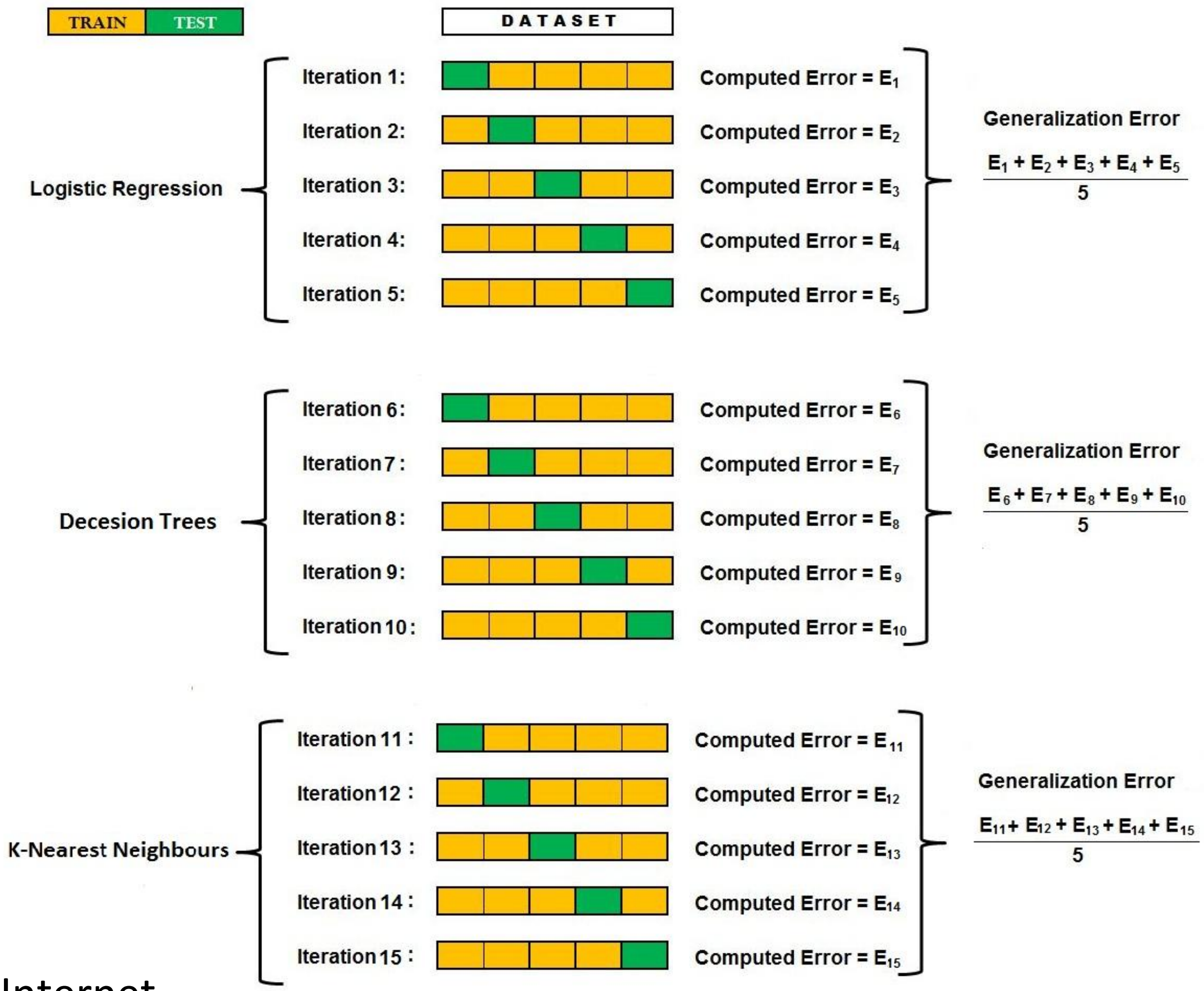
$$J_{\text{test}}(\theta) = \frac{1}{2m_{\text{test}}} \sum_{i=1}^{\text{test}} \left(h_\theta(x_{\text{test}}^{(i)}) - y_{\text{test}}^{(i)} \right)^2$$

k-fold cross validation

- Randomly split dataset into k parts
- Learn and validate k times. k is usually 10



k-fold cross validation

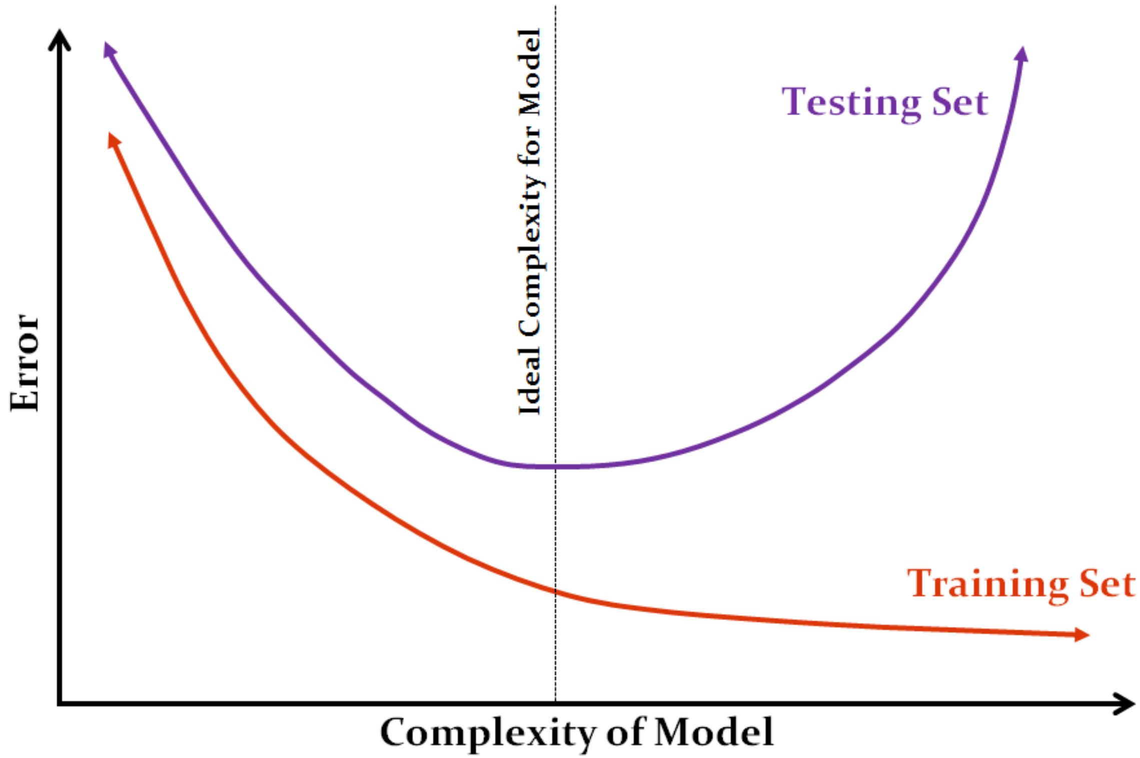


Source: Internet

Model finetuning

- Model has various parameters
- Learning algorithm has various parameters as well, namely hyper parameters
- What should we do when model does not work well?
  - Collect more data
  - Reduce number of features
  - Try with new features
  - Switch to other models or hypotheses
  - Reduce weight decay
  - Increase weight decay

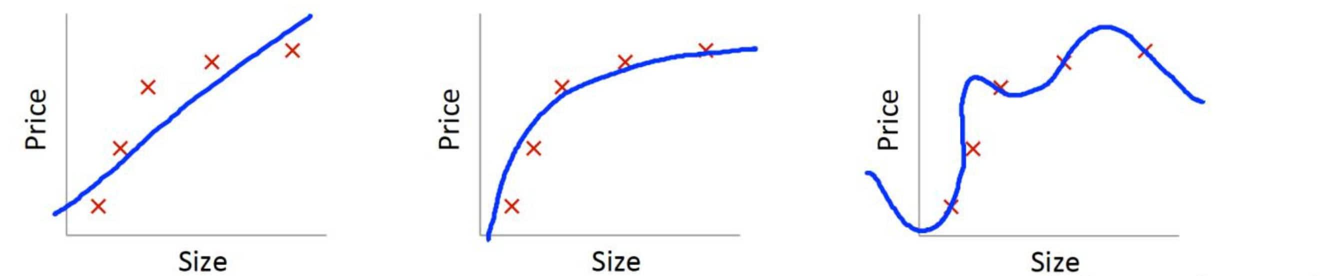
Model finetuning



Source: Internet

Bias and variance

- Bias: model error on training set
- Variance: difference between model error on validation set and that on training set
- Adjust bias and variance until they reach minimal

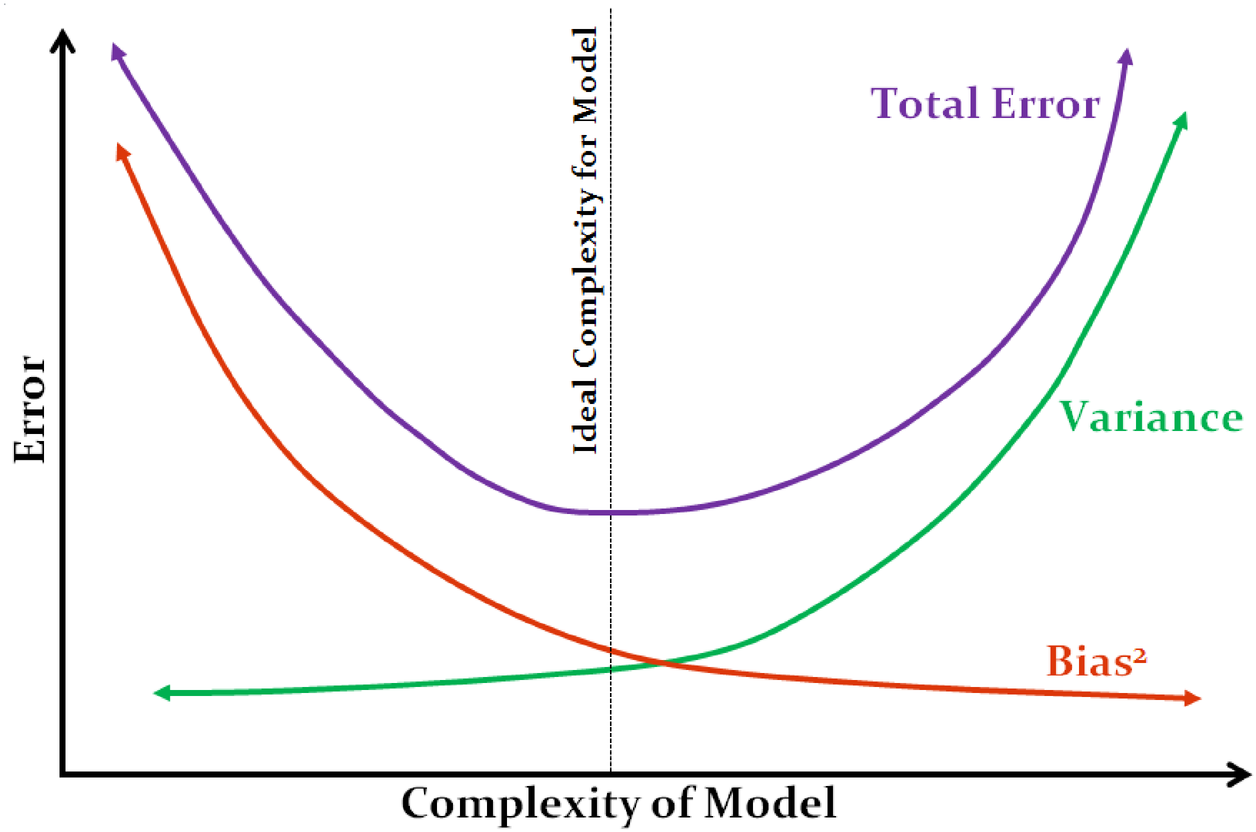


$\theta_0 + \theta_1 x$ | $\theta_0 + \theta_1 x + \theta_2 x^2$ | $\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

High bias (underfit) | "Just right" | High variance (overfit)

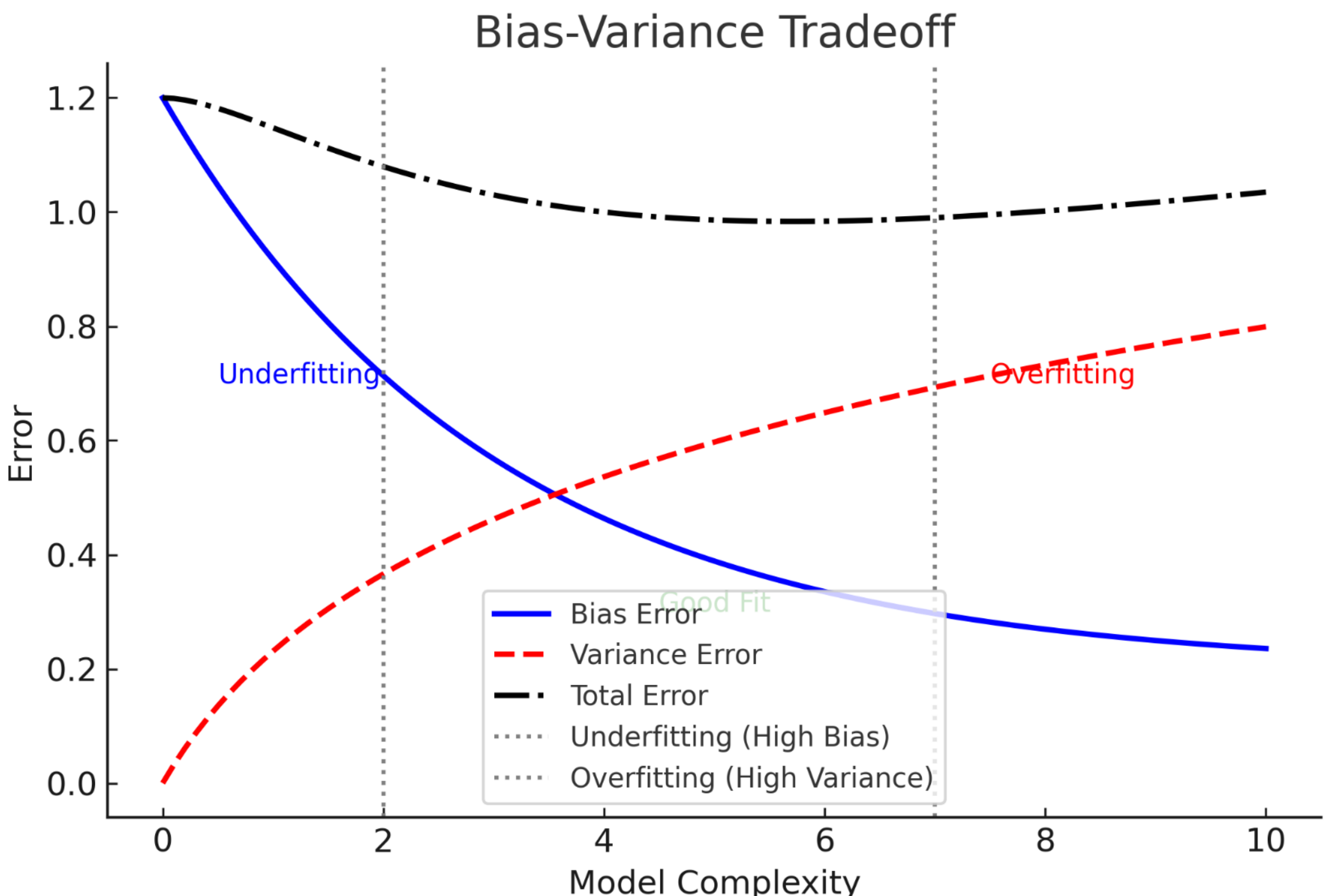
Source: Andrew Ng

Bias and variance



Source: Internet

Bias and variance



Source: ChatGPT

Model evaluation

- Measurements
  - Precision, recall
  - Accuracy
  - F1-score
- Depending on problems, we need some suitable measures

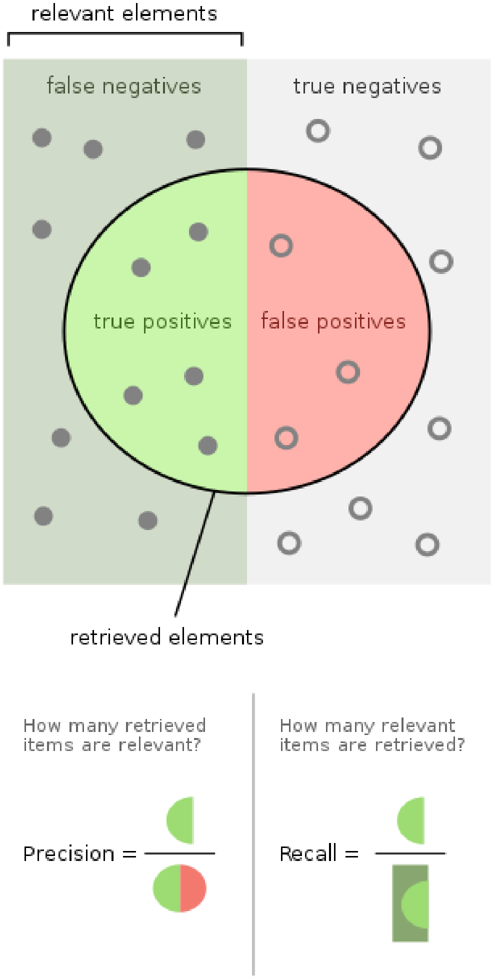
Model evaluation

- $$\text{Precision} = \frac{\text{True positive}}{\text{Predicted positive}}$$

- $$\text{Recall} = \frac{\text{True positive}}{\text{Positive}}$$

- $$\text{F1} - \text{score} = \frac{2 \text{ x Precision x Recall}}{\text{Precision} + \text{Recall}}$$

- $$\text{Accuracy} = \frac{\text{True positive} + \text{True negative}}{\text{Positive} + \text{Negative}}$$



Source: Wikipedia

Confusion matrix

		Predicted				Total
		Cat	Dog	Tiger	Wolf	
Actual	Cat	**6**	0	3	1	10
	Dog	2	**4**	0	4	10
	Tiger	3	3	**3**	0	9
	Wolf	1	4	1	**2**	8
Total		12	11	7	7	

>>> sklearn.metrics.classification_report

```
>>> y_true =  [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …]
>>> y_pred = [0, 0, 0, 0, 0, 0, 2, 2, 2, 3, …]
>>> target_names = [‘Cat', ‘Dog', ‘Tiger’, ‘Wolf’]
```

```
              precision    recall  f1-score   support

         Cat      0.500     0.600     0.545        10
         Dog      0.364     0.400     0.381        10
       Tiger      0.429     0.333     0.375         9
        Wolf      0.286     0.250     0.267         8

    accuracy                          0.405        37
   macro avg      0.394     0.396     0.392        37
weighted avg      0.399     0.405     0.399        37
```

Confusion matrix

- Example code

```
from sklearn.metrics import classification_report

y_true = [0, 0, 1, 1, 2, 2]
y_pred = [0, 1, 1, 1, 2, 0]
target_names = ["Class A", "Class B", "Class C"]

print(classification_report(y_true, y_pred, target_names=target_names))
```

Confusion matrix

		Predicted				Total
		Cat	Dog	Tiger	Wolf	
Actual	Cat	**6**	0	3	1	10
	Dog	2	**4**	0	4	10
	Tiger	3	3	**3**	0	9
	Wolf	1	4	1	**2**	8
Total		12	11	7	7	

- Macro average is calculated over all classes regardless their proportions
- Weighted average is calculated over all classes taking their proportions into account

	precision	recall	f1-score	support
Cat	0.500	0.600	0.545	10
Dog	0.364	0.400	0.381	10
Tiger	0.429	0.333	0.375	9
Wolf	0.286	0.250	0.267	8
accuracy			0.405	37
macro avg	0.394	0.396	0.392	37
weighted avg	0.399	0.405	0.399	37

Quiz

- 1. Which of the following scenarios indicates an overfitting model?
  - A) Low training loss, high test loss
  - B) High training loss, low test loss
  - C) Both training and test loss are high
  - D) Both training and test loss are low
- 2. What is the primary purpose of L2 regularization in machine learning?
  - A) Encourages sparsity by setting some weights to zero
  - B) Penalizes large weights to prevent overfitting
  - C) Speeds up training without affecting model complexity
  - D) Removes unnecessary features from the dataset

Quiz

- 3. In early stopping, at which point should you stop training?
  - A) When training loss is minimized
  - B) When validation loss starts increasing
  - C) When both training and validation loss reach zero
  - D) After a fixed number of epochs, regardless of loss behavior
- 4. Which technique helps evaluate a model's generalization ability without using separate training and test sets?
  - A) Dropout Regularization
  - B) L1/L2 Regularization
  - C) k-Fold Cross-Validation
  - D) Batch Normalization

Exercise

		Predicted				Total
		Cat	Dog	Tiger	Wolf	
Actual	Cat	**6**	0	3	1	10
	Dog	2	**4**	0	4	10
	Tiger	3	3	**3**	0	9
	Wolf	1	4	1	**2**	8
	Total	12	11	7	7	

- Calculate precision, recall and F1-score of class dog

Exercise

		Predicted				Total
		Cat	Dog	Tiger	Wolf	
Actual	Cat	**6**	0	3	1	10
	Dog	2	**4**	0	4	10
	Tiger	3	3	**3**	0	9
	Wolf	1	4	1	**2**	8
Total		12	11	7	7	

- Calculate macro average and weighted average of precision